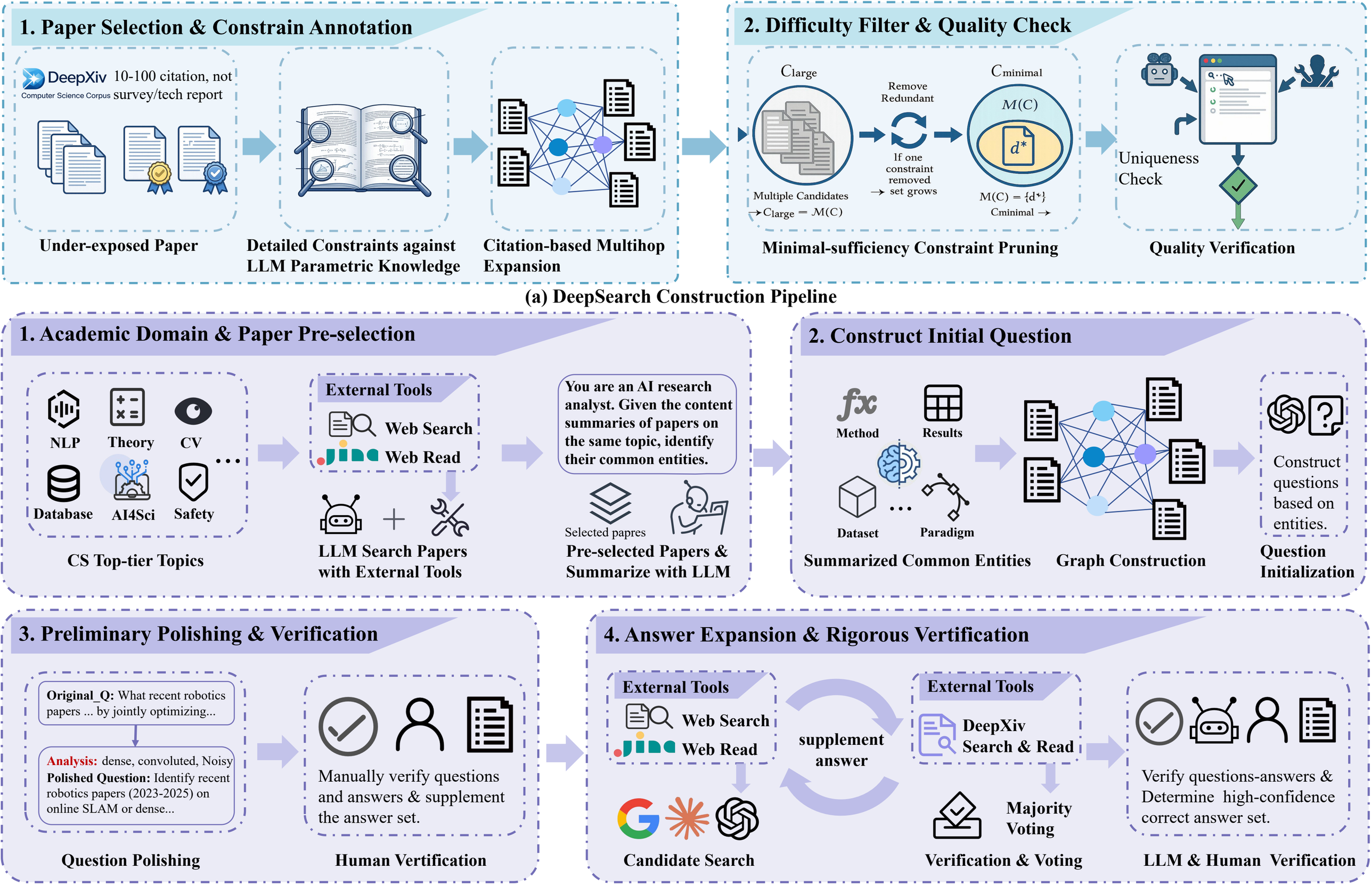
1. Paper Selection & Constrain Annotation
DeepXiv
Computer Science Corpus
10-100 citation, not survey/tech report
Under-exposed Paper
Detailed Constraints against LLM Parametric Knowledge
Citation-based Multihop Expansion
2. Difficulty Filter & Quality Check
Clarge
Multiple Candidates
→ Clarge = M(C)
Remove Redundant
If one constraint removed → set grows
Cminimal
M(C)
d*
M(C) = {d*}
Cminimal →
Minimal-sufficiency Constraint Pruning
Uniqueness Check
Quality Verification
(a) DeepSearch Construction Pipeline
1. Academic Domain & Paper Pre-selection
NLP
Theory
CV
Database
AI4Sci
Safety
CS Top-tier Topics
External Tools
Web Search
Web Read
LLM Search Papers with External Tools
You are an AI research analyst. Given the content summaries of papers on the same topic, identify their common entities.
Selected papres
Pre-selected Papers & Summarize with LLM
2. Construct Initial Question
Method
Results
Dataset
Paradigm
Summarized Common Entities
Graph Construction
Construct questions based on entities.
Question Initialization
3. Preliminary Polishing & Verification
Original_Q: What recent robotics papers ... by jointly optimizing...
Analysis: dense, convoluted, Noisy
Polished Question: Identify recent robotics papers (2023-2025) on online SLAM or dense...
Question Polishing
Manually verify questions and answers & supplement the answer set.
Human Vertification
4. Answer Expansion & Rigorous Vertification
External Tools
Web Search
Web Read
Candidate Search
supplement answer
External Tools
DeepXiv Search & Read
Majority Voting
Verification & Voting
Verify questions-answers & Determine high-confidence correct answer set.
LLM & Human Verification
(b) WideSearch Construction Pipeline